AGE 310: Instrumentation and Measurements

Dr. Kadeghe Fue, Department of Agricultural Engineering, SUA

Morogoro, Tanzania

Abstract

Lecture 5 examines active transducers, focusing on two key mechanisms: the Piezoelectric Effect and the Photoelectric Effect. We detail how mechanical energy (force, acceleration) and light energy are converted into electrical signals. This knowledge is vital for understanding high-frequency measurement applications like vibration monitoring and optical sensing used in machine vision systems for quality control and field automation.

Lecture 5: Piezoelectric and Photoelectric Transducers

5.1 Piezoelectric Transducers (Active)

5.1.1 The Piezoelectric Effect

Piezoelectricity (from Greek *piezein*, meaning to squeeze or press) is the ability of certain crystalline materials to generate an electric charge in response to applied mechanical stress.

- **Active Transducer:** Piezoelectric devices are active because they **generate their own charge** (voltage) and do not require an external power supply for the measurement itself.
- **Materials:** Commonly used materials include naturally occurring Quartz (SiO_2) and synthetic ceramics like Lead Zirconate Titanate (PZT). PZT offers higher sensitivity but is less stable than Quartz.
- **Reversibility:** The effect is reversible: applying an electric field causes the material to change shape (used in ultrasonic emitters and buzzers).

The generated charge (Q) is directly proportional to the applied force (F):

$$Q = d \cdot F$$

Where d is the piezoelectric charge constant (C/N).

5.1.2 Application: Accelerometers and Load Cells

Piezoelectric materials are ideally suited for measuring dynamic quantities (rapid changes in force or motion).

- **Accelerometers:** A mass (seismic mass) is placed on the crystal. When the sensor accelerates (a), the mass exerts a force ($F = ma$) on the crystal, generating a charge proportional to the acceleration. This is critical for monitoring tractor ride quality and vibration in engine components.
- **Knock Sensors:** Mounted on tractor engine blocks, they detect the characteristic high-frequency pressure waves (knock) caused by premature combustion, allowing the ECU to adjust ignition timing.
- **Dynamic Force Measurement:** Used in research to measure impact forces (e.g., fruit bruising, soil cutting forces).

5.1.3 Limitations

- **No Static Measurement:** Piezoelectric devices cannot measure constant (static) quantities (like dead weight or steady pressure). The accumulated charge quickly leaks away due to the inherent internal impedance of the crystal and the external circuit, leading to measurement drift.
- **High Output Impedance:** Requires specialized charge amplifiers (signal conditioning) to boost the signal without corrupting the measurement or experiencing significant signal loss.

Calculation Example 1: Charge Generation

A Piezoelectric force sensor uses PZT with a charge constant $d = 300$ pC/N (picoCoulombs per Newton
If the sensor is subjected to a transient impact force of 5 kN (5000 N), calculate the resulting electric charge generated.

Solution:

$$Q = d \cdot F$$

$$Q = (300 \times 10^{-12} \text{ C/N}) \times 5000 \text{ N}$$

$$Q = 1.5 \times 10^{-6} \text{ C} = 1.5 \text{ μC}$$

This microcoulomb charge is then processed by a charge amplifier to produce a voltage output.

5.2 Photoelectric Transducers

These transducers exploit the interaction of light (photons) with semiconductor materials to produce a measurable electrical signal. They are essential for non-contact object detection and light intensity measurement.

5.2.1 Light Dependent Resistor (LDR)

- **Principle (Photoconductive):** Made from a semiconductor material (e.g., Cadmium Sulfide, CdS).
- **Operation:** In the absence of light, the material has high resistance. When photons strike the material, they excite electrons into the conduction band, increasing conductivity. **Resistance decreases** as light intensity increases.
- **Type:** Passive transducer (requires external voltage to measure current/resistance).
- **Application:** Simple light switches, automatic controls for irrigation lights.
- **Limitation:** Very slow response time (milliseconds or seconds), unsuitable for fast counting.

5.2.2 Photodiodes and Phototransistors

These devices are based on the semiconductor P-N junction.

- **Photodiode:** Operates in **reverse bias**. Light striking the junction generates an electron-hole pair, creating a small reverse current ($\propto$ light intensity).
  - **Pros:** Very fast response (nanoseconds), highly linear.
  - **Cons:** Low current output, requires amplification.
- **Phototransistor:** Similar to a photodiode, but the generated current is internally amplified by transistor action (the base current is proportional to light).
  - **Pros:** High sensitivity, large current output.
  - **Cons:** Slower response than a photodiode.

5.2.3 Photovoltaic Cells (Active)

- **Principle:** Light energy is directly converted into electrical energy (voltage/current).
- **Type:** Active transducer (no external power required).
- **Applications:** Used as the main power source for many remote IoT weather stations (SUA EPA lab). When used as a sensor (Pyranometer), the short-circuit current is used to measure solar radiation intensity (W/m^2) for calculating crop evapotranspiration.

5.3 Application: Optical Encoders and Counting

5.3.1 Optical Encoders

Optical encoders use light to measure rotational speed (RPM) or linear position precisely.

- **Construction:** Consists of a light source (LED), a slotted disk (or strip), and a photodetector (photodiode or phototransistor).
- **Operation:** As the disk rotates, the slots interrupt the light beam, generating a pulse train. Counting the pulses over time gives the speed.
- **Types:**
  - *Incremental:* Measures change in position (relative).
  - *Absolute:* Measures the exact position (absolute) using unique coded patterns on the disc.

5.3.2 Seed Monitoring in Planters

- **Problem:** Ensuring the correct seed spacing (*singulation*) and population (seeds per hectare).
- **Solution:** Optical sensors are placed inside the seed tube of a precision planter. Each time a seed passes, it breaks the LED beam.
- **Function:** The microcontroller counts the pulses to verify the planter's performance (skips, doubles, population). This is a Zero Order counting process.

5.4 Summary of Key Terms (Swahili/Context)

- **Active Transducer:** Kifaa kinachotoa nishati (Generates its own electrical signal).
- **Piezoelectric Effect:** Athari ya Piezoelectric (Generating charge from pressure/force).
- **LDR (Light Dependent Resistor):** Kipinga kutegemea Mwanga (Resistance changes with light).
- **Photoconductive:** Kuongeza uwezo wa kupitisha umeme kwa mwanga (Resistance drops when illuminated).
- **Optical Encoder:** Kichunguzi cha Mwanga (Measures position or speed using light interruption).
- **VLOS:** Kuona kwa macho (Visual Line of Sight - relevant for drone operation).

Lecture 5: Review Questions

SECTION A: Basic Concepts (5 Questions)

1. Define the Piezoelectric Effect. Is a piezoelectric transducer an active or passive device, and why?
2. State the primary limitation of using a piezoelectric sensor for measuring the static weight of grain in a storage container.
3. Differentiate between an LDR (Light Dependent Resistor) and a Phototransistor based on (a) device type (passive/active) and (b) speed of response.
4. Explain how an Optical Encoder uses light to measure the RPM of a rotating shaft.
5. Briefly describe the function of a Pyranometer (a solar radiation sensor). Which photoelectric principle does it employ?

SECTION B: Advanced Analysis and Calculations (10 Questions)

6. **Accelerometer Charge Amplifier:** Explain why the output of a Piezoelectric Accelerometer must be fed into a high-impedance *Charge Amplifier* rather than a standard Op-Amp voltage amplifier. What happens to the measurement if a standard voltage amplifier is used?
7. **Sensitivity of Photodiode:** A photodiode has a responsivity of 0.5 A/W at 600 nm wavelength. If the incident light power is 200 µW (200×10^{-6} W), calculate the resulting current output. If this current is fed into a 100 kΩ feedback resistor of a Transimpedance Amplifier, what is the output voltage?
8. **LDR Dynamic Range:** An LDR used for ambient light measurement has a resistance of 1 MΩ in the dark and 1 kΩ in full sunlight. If the LDR is part of a voltage divider circuit with a fixed 10 kΩ resistor and a 5 V source, calculate the maximum and minimum output voltages.
9. **Optical Encoder Resolution:** A rotary incremental optical encoder has 500 slots on its disc. The rotating shaft reaches a maximum speed of 1200 RPM.
    - Calculate the angular resolution in degrees per pulse.
    - Calculate the maximum output pulse frequency (in Hz).
10. **Tractor Engine Knock:** Explain the role of the Piezoelectric Knock Sensor in maintaining tractor engine efficiency and longevity. What adjustment does the Engine Control Unit (ECU) make when it detects a knock signal?
11. **Spectral Filtering in Vision:** When using a camera system for automated fruit harvesting, why might an engineer choose to use a red or near-infrared (NIR) filter instead of a standard RGB camera? Relate your answer to the spectral properties of ripe fruit and green leaves.
12. **Trade-offs in Counting:** An automated seed counter requires high-speed response (up to 1000 seeds per second). Would you select an LDR or a Photodiode for the detection element, and why?

13. **Phototransistor vs Photodiode:** Why might a Phototransistor be preferred over a Photodiode in a basic object detection application where the distance between the LED and the detector is large, despite the Photodiode being faster?

14. **Powering Remote Sensors:** A remote IoT weather station uses a photovoltaic cell for primary power. If the cell produces 100 mW of power when exposed to sunlight, and the IoT module consumes 20 mW on average, calculate the maximum operating duration of the system's 1000 µAh buffer battery before it is depleted. *Assume the battery is initially fully charged.*

15. **Engine Knock Mitigation (Acoustics):** The Piezoelectric sensor detects engine knock. Explain why a sophisticated signal processing algorithm (filtering) is required to isolate the destructive knock frequency from normal engine vibration and noise. What frequency range is typically monitored for knock detection?